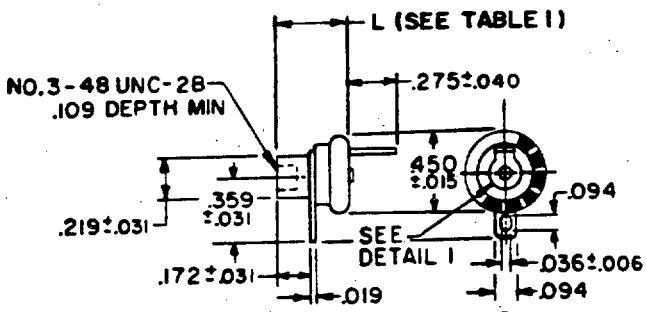
MIL-C-10950/2D
8 July 1968
SUPERSEDING
MIL-C-10950/2C
7 October 1963

MILITARY SPECIFICATION SHEET

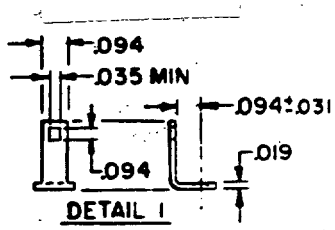
CAPACITORS, FIXED, MICA DIELECTRIC, BUTTON STYLE (STANDOFF) STYLE CB21

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-10950.

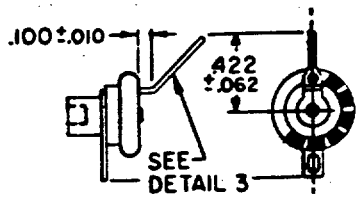
This specification is mandatory for use by all Departments and Agencies of the Department of Defense.



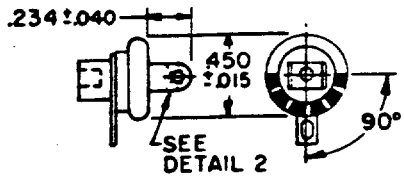
WITH TERMINAL ASSEMBLY P AND SHELL DIMENSIONS



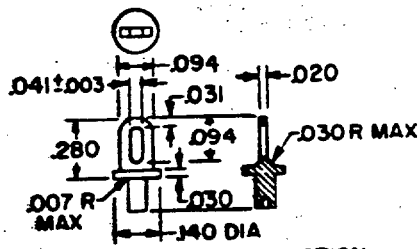
DETAIL I



WITH TERMINAL ASSEMBLY T



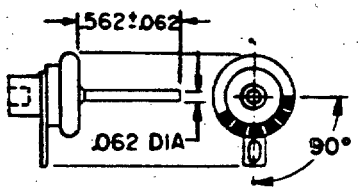
WITH TERMINAL ASSEMBLY Q



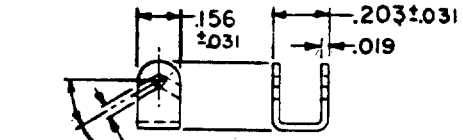
OPTIONAL CONSTRUCTION FOR DETAIL I

FSC 5910

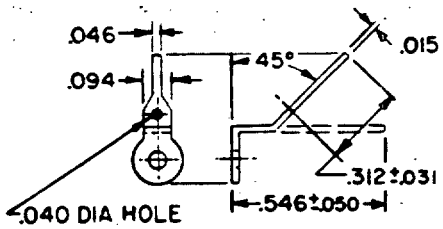
Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:38Z
Check the source to verify that this is the current version before use.



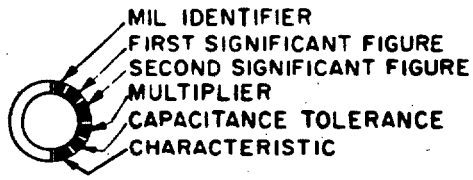
WITH TERMINAL ASSEMBLY V



DETAIL 2



DETAIL 3



COLOR-CODE MARKING

INCHES	MM
.003	.08
.005	.13
.006	.15
.007	.18
.010	.25
.015	.38
.019	.48
.020	.51
.031	.79
.035	.89
.036	.91
.040	1.02
.046	1.17
.047	1.19
.050	1.27
.062	1.57
.094	2.39
.100	2.54
.109	2.77
.140	3.56
.156	3.96
.172	4.37
.203	5.16
.219	5.56
.234	5.94
.275	6.99
.280	7.11
.312	7.92
.359	9.12
.380	9.65
.422	10.72
.438	11.13
.450	11.43
.546	13.87
.562	14.27

NOTES:
1. All dimensions in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are .xxx±.005 (.13 mm) and ±15° on angles.
4. All dimensions are exclusive of sealing compound and color code.
5. Color-coded section shall not exceed 240°.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:38Z
Check the source to verify that this is the current version before use.

TABLE I. Style CB21.

Part number 1/	Capacitance	DC rated voltage	L Dimension ±.040	Characteristics available at capacitance tolerance				Terminal assemblies
				±1% (F)	±2% (G)	±5% (J)	±10% (K)	
	pf	Volts						
CB21-B050K	5	500	0.380				B	P,Q,T,V
CB21-D100K	10	500	0.380				D	P,Q,T,V
CB21-D150K	15	500	0.380				D	P,Q,T,V
CB21-D180K	18	500	0.380				D	P,Q,T,V
CB21-D200J	20	500	0.380			D	-	P,Q,T,V
CB21-D220-	22	500	0.380			D	D	P,Q,T,V
CB21-D240J	24	500	0.380			D	-	P,Q,T,V
CB21-D270-	27	500	0.380			D	D	P,Q,T,V
CB21-D300J	30	500	0.380			D	-	P,Q,T,V
CB21-D330-	33	500	0.380			D	D	P,Q,T,V
CB21-D390-	39	500	0.380			D	D	P,Q,T,V
CB21-D470-	47	500	0.380			D	D	P,Q,T,V
CB21--510-	51	500	0.380	D,E	D,E	D,E	-	P,Q,T,V
CB21--560-	56	500	0.380	D,E	D,E	D,E	D	P,Q,T,V
CB21--620-	62	500	0.380	D,E	D,E	D,E	-	P,Q,T,V
CB21--680-	68	500	0.380	D,E	D,E	D,E	D	P,Q,T,V
CB21--750-	75	500	0.380	D,E	D,E	D,E	D	P,Q,T,V
CB21--101-	100	500	0.380	D,E	D,E	D,E	D	P,Q,T,V
CB21--111-	110	500	0.380	D,E	D,E	D,E	D	P,Q,T,V
CB21--151-	150	500	0.380	D,E	D,E	D,E	D	P,Q,T,V
CB21--161-	160	500	0.380	D,E	D,E	D,E	D	P,Q,T,V
CB21--201-	200	500	0.380	D,E	D,E	D,E	D	P,Q,T,V
CB21--221-	220	500	0.380	D,E	D,E	D,E	-	P,Q,T,V
CB21--241-	240	500	0.380	D,E	D,E	D,E	D	P,Q,T,V
CB21--271-	270	500	0.380	D,E	D,E	D,E	-	P,Q,T,V
CB21--301-	300	500	0.380	D,E	D,E	D,E	D	P,Q,T,V
CB21--331-	330	500	0.380	D,E	D,E	D,E	D	P,Q,T,V
CB21--361-	360	500	0.380	D,E	D,E	D,E	D	P,Q,T,V
CB21--431-	430	500	0.380	D,E	D,E	D,E	D	P,Q,T,V
CB21--471-	470	500	0.380	D,E	D,E	D,E	-	P,Q,T,V
CB21--511-	510	500	0.380	D,E	D,E	D,E	D	P,Q,T,V
CB21--561-	560	500	0.380	D,E	D,E	D,E	-	P,Q,T,V
CB21--621-	620	500	0.380	D,E	D,E	D,E	D	P,Q,T,V
CB21--681-	680	500	0.380	D,E	D,E	D,E	D	P,Q,T,V
CB21--751-	750	500	0.380	D,E	D,E	D,E	D	P,Q,T,V
CB21--102-	1,000	500	0.380	D,E	D,E	D,E	D	P,Q,T,V
CB21--152-	1,500	500	0.438	D,E	D,E	D,E	D	P,Q,T,V
CB21--182	1,800	500	0.438	D,E	D,E	D,E	D	P,Q,T,V
CB21--222-	2,200	500	0.438	D,E	D,E	D,E	D	P,Q,T,V
CB21--242-	2,400	500	0.438	D,E	D,E	D,E	D	P,Q,T,V

1/ Complete part number will include additional symbols to indicate terminal assembly, characteristic, and capacitance tolerance.

REQUIREMENTS:

Dimensions and configuration:

Case: See figure and table I.

Terminals: See figure and table I.

Capacitance value: See table I.

Capacitance tolerance: See table I.

DC rated voltage: 500 volts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:38Z
Check the source to verify that this is the current version before use.

Dissipation factor: See figure 1 of MIL-C-10950.

Characteristics: See table I.

Symbol	Temperature coefficient	Capacitance drift
B	Not specified	Not specified
D	-100 to +100	0.3 percent or 0.3 pf whichever is greater
E	-20 to +100	± (0.1 percent + 0.1 pf)

Operating temperature range: -55° C to 85° C.

Dielectric withstanding voltage:
Method 301 of MIL-STD-202:
200 percent of rated dc voltage for not less than 1 second, nor more than 5 seconds.

Barometric pressure:
Method 105 of MIL-STD-202, condition C (70, 000 ft.)
Test potential - 125 percent of rated dc voltage for not less than 1 second nor more than 5 seconds. Surge current shall not exceed 5 ma.

Insulation resistance:
Method 302 of MIL-STD-202, condition A or B.
At 25° C - 50, 000 megohms minimum.
At 85° C - 15, 000 megohms minimum.

Capacitance: Method 305 of MIL-STD-202.
Measurement frequency
Capacitance values of 1000 pf or less - 1MHz ±100 kHz.
Capacitance values of greater than 1000 pf - 1kHz ±100 Hz.

Shock (medium impact):
Method 205 of MIL-STD-202, condition C (50G).

Vibration: Method 204 of MIL-STD-202, condition B.

Moisture resistance: Test method 106 of MIL-STD-202.
Insulation resistance:
Not less than 500 megohms:
Capacitance change:
Within 1 percent of the initial value or 0.5 pf, whichever is greater.
DF - 200 percent of initial requirement.

Temperature cycling and immersion:
Temperature cycling:
Test method 102 of MIL-STD-202, condition D.
Immersion:
Test method 104 of MIL-STD-202, condition A.
Insulation resistance:
Not less than 5000 megohms.
Capacitance change:
Within 1 percent of the initial value or 0.5 pf, whichever is greater.

Life:
2000 ± 12 hours at 85° $^{+3°}_{-0°}$ C.
Insulation resistance:
Greater than 50, 000 megohms.
Capacitance change:
Within 3 percent of the initial value or 0.5 pf, whichever is greater.

Temperature coefficient and capacitance drift:
Within the limits specified for the applicable characteristic.

Torque:
In accordance with MIL-C-10950.
Mounting - Normal mounting means.
Load - 4 ounce-inch.

NOTES:
1. Conversion of dash numbers to part numbers is as shown in table II.
2. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:38Z
Check the source to verify that this is the current version before use.

TABLE II. Dash number to part number conversion.

Dash number in MIL-C-10950/2C								Converted to part number 1/ in MIL-C-10950/2D
Characteristic D				Characteristic E				
Terminal assembly				Terminal assembly				
P	Q	T	V	P	Q	T	V	
0001	0002	0003	0004	0005	0006	0007	0008	CB21--510F
0009	0010	0011	0012	0013	0014	0015	0016	CB21--560F
0017	0018	0019	0020	0021	0022	0023	0024	CB21--620F
0025	0026	0027	0028	0029	0030	0031	0032	CB21--680F
0033	0034	0035	0036	0037	0038	0039	0040	CB21--750F
0041	0042	0043	0044	0045	0046	0047	0048	CB21--101F
0049	0050	0051	0052	0053	0054	0055	0056	CB21--111F
0057	0058	0059	0060	0061	0062	0063	0064	CB21--151F
0065	0066	0067	0068	0069	0070	0071	0072	CB21--161F
0073	0074	0075	0076	0077	0078	0079	0080	CB21--201F
0081	0082	0083	0084	0085	0086	0087	0088	CB21--221F
0089	0090	0091	0092	0093	0094	0095	0096	CB21--241F
0097	0098	0099	0100	0101	0102	0103	0104	CB21--271F
0105	0106	0107	0108	0109	0110	0111	0112	CB21--301F
0113	0114	0115	0116	0117	0118	0119	0120	CB21--331F
0121	0122	0123	0124	0125	0126	0127	0128	CB21--361F
0129	0130	0131	0132	0133	0134	0135	0136	CB21--431F
0137	0138	0139	0140	0141	0142	0143	0144	CB21--471F
0145	0146	0147	0148	0149	0150	0151	0152	CB21--511F
0153	0154	0155	0156	0157	0158	0159	0160	CB21--561F
0161	0162	0163	0164	0165	0166	0167	0168	CB21--621F
0169	0170	0171	0172	0173	0174	0175	0176	CB21--681F
0177	0178	0179	0180	0181	0182	0183	0184	CB21--751F
0185	0186	0187	0188	0189	0190	0191	0192	CB21--102F
0193	0194	0195	0196	0197	0198	0199	0200	CB21--152F
0201	0202	0203	0204	0205	0206	0207	0208	CB21--182F
0209	0210	0211	0212	0213	0214	0215	0216	CB21--222F
0217	0218	0219	0220	0221	0222	0223	0224	CB21--242F
0225	0226	0227	0228	0229	0230	0231	0232	CB21--510G
0233	0234	0235	0236	0237	0238	0239	0240	CB21--560G
0241	0242	0243	0244	0245	0246	0247	0248	CB21--620G
0249	0250	0251	0252	0253	0254	0255	0256	CB21--680G
0257	0258	0259	0260	0261	0262	0263	0264	CB21--750G
0265	0266	0267	0268	0269	0270	0271	0272	CB21--101G
0273	0274	0275	0276	0277	0278	0279	0280	CB21--111G

1/ See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:38Z
Check the source to verify that this is the current version before use.

TABLE II. Dash number to part number conversion (continued).

Dash number in MIL-C-10950/2C								Converted to part number 1/ in MIL-C-10950/2D
Characteristic D				Characteristic E				
Terminal assembly				Terminal assembly				
P	Q	T	V	P	Q	T	V	
0281	0282	0283	0284	0285	0286	0287	0288	CB21--151G
0289	0290	0291	0292	0293	0294	0295	0296	CB21--161G
0297	0298	0299	0300	0301	0302	0303	0304	CB21--201G
0305	0306	0307	0308	0309	0310	0311	0312	CB21--221G
0313	0314	0315	0316	0317	0318	0319	0320	CB21--241G
0321	0322	0323	0324	0325	0326	0327	0328	CB21--271G
0329	0330	0331	0332	0333	0334	0335	0336	CB21--301G
0337	0338	0339	0340	0341	0342	0343	0344	CB21--331G
0345	0346	0347	0348	0349	0350	0351	0352	CB21--361G
0353	0354	0355	0356	0357	0358	0359	0360	CB21--431G
0361	0362	0363	0364	0365	0366	0367	0368	CB21--471G
0369	0370	0371	0372	0373	0374	0375	0376	CB21--511G
0377	0378	0379	0380	0381	0382	0383	0384	CB21--561G
0385	0386	0387	0388	0389	0390	0391	0392	CB21--621G
0393	0394	0395	0396	0397	0398	0399	0400	CB21--681G
0401	0402	0403	0404	0405	0406	0407	0408	CB21--751G
0409	0410	0411	0412	0413	0414	0415	0416	CB21--102G
0417	0418	0419	0420	0421	0422	0423	0424	CB21--152G
0425	0426	0427	0428	0429	0430	0431	0432	CB21--182G
0433	0434	0435	0436	0437	0438	0439	0440	CB21--222G
0441	0442	0443	0444	0445	0446	0447	0448	CB21--242G
0449	0450	0451	0452					CB21--200J
0453	0454	0455	0456					CB21--220J
0457	0458	0459	0460					CB21--240J
0461	0462	0463	0464					CB21--270J
0465	0466	0467	0468					CB21--300J
0469	0470	0471	0472					CB21--330J
0473	0474	0475	0476					CB21--390J
0477	0478	0479	0480					CB21--470J
0481	0482	0483	0484	0485	0486	0487	0488	CB21--510J
0489	0490	0491	0492	0493	0494	0495	0496	CB21--560J
0497	0498	0499	0500	0501	0502	0503	0504	CB21--620J
0505	0506	0507	0508	0509	0510	0511	0512	CB21--680J
0513	0514	0515	0516	0517	0518	0519	0520	CB21--750J
0521	0522	0523	0524	0525	0526	0527	0528	CB21--101J

1/ See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:38Z
Check the source to verify that this is the current version before use.

TABLE II. Dash number to part number conversion (continued).

Dash number in MIL-C-10950/2C								Converted to part number 1/ in MIL-C-10950/2D
Characteristic D				Characteristic E				
Terminal assembly				Terminal assembly				
P	Q	T	V	P	Q	T	V	
0529	0530	0531	0532	0533	0534	0535	0536	CB21--111J
0537	0538	0539	0540	0541	0542	0543	0544	CB21--151J
0545	0546	0547	0548	0549	0550	0551	0552	CB21--161J
0553	0554	0555	0556	0557	0558	0559	0560	CB21--201J
0561	0562	0563	0564	0565	0566	0567	0568	CB21--221J
0569	0570	0571	0572	0573	0574	0575	0576	CB21--241J
0577	0578	0579	0580	0581	0582	0583	0584	CB21--271J
0585	0586	0587	0588	0589	0590	0591	0592	CB21--301J
0593	0594	0595	0596	0597	0598	0599	0600	CB21--331J
0601	0602	0603	0604	0605	0606	0607	0608	CB21--361J
0609	0610	0611	0612	0613	0614	0615	0616	CB21--431J
0617	0618	0619	0620	0621	0622	0623	0624	CB21--471J
0625	0626	0627	0628	0629	0630	0631	0632	CB21--511J
0633	0634	0635	0636	0637	0638	0639	0640	CB21--561J
0641	0642	0643	0644	0645	0646	0647	0648	CB21--621J
0649	0650	0651	0652	0653	0654	0655	0656	CB21--681J
0657	0658	0659	0660	0661	0662	0663	0664	CB21--751J
0665	0666	0667	0668	0669	0670	0671	0672	CB21--102J
0673	0674	0675	0676	0677	0678	0679	0680	CB21--152J
0681	0682	0683	0684	0685	0686	0687	0688	CB21--182J
0689	0690	0691	0692	0693	0694	0695	0696	CB21--222J
0697	0698	0699	0700	0701	0702	0703	0704	CB21--242J
Characteristic B				Characteristic D				
0705	0706	0707	0708					CB21-B050K
				0709	0710	0711	0712	CB21-D100K
				0713	0714	0715	0716	CB21-D150K
				0717	0718	0719	0720	CB21-D180K
				0721	0722	0723	0724	CB21-D220K
				0725	0726	0727	0728	CB21-D270K
				0729	0730	0731	0732	CB21-D330K
				0733	0734	0735	0736	CB21-D390K
				0737	0738	0739	0740	CB21-D470K
				0741	0742	0743	0744	CB21-D560K

1/ See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:38Z
Check the source to verify that this is the current version before use.

TABLE II. Dash number to part number conversion (continued).

Dash number in MIL-C-10950/2C								Converted to part number 1/ in MIL-C-10950/2D
Characteristic B				Characteristic D				
Terminal assembly				Terminal assembly				
P	Q	T	V	P	Q	T	V	
				0745	0746	0747	0748	CB21-D680K
				0749	0750	0751	0752	CB21-D750K
				0753	0754	0755	0756	CB21-D101K
				0757	0758	0759	0760	CB21-D111K
				0761	0762	0763	0764	CB21-D151K
				0765	0766	0767	0768	CB21-D161K
				0769	0770	0771	0772	CB21-D201K
				0773	0774	0775	0776	CB21-D221K
				0777	0778	0779	0780	CB21-D271K
				0781	0782	0783	0784	CB21-D331K
				0785	0786	0787	0788	CB21-D361K
				0789	0790	0791	0792	CB21-D431K
				0793	0794	0795	0796	CB21-D471K
				0797	0798	0799	0800	CB21-D561K
				0801	0802	0803	0804	CB21-D681K
				0805	0806	0807	0808	CB21-D751K
				0809	0810	0811	0812	CB21-D102K
				0813	0814	0815	0816	CB21-D152K
				0817	0818	0819	0820	CB21-D182K
				0821	0822	0823	0824	CB21-D222K
				0825	0826	0827	0828	CB21-D242K

1/ The complete part number shall include symbols for terminal assembly and characteristic, as applicable.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU
Navy - SH
Air Force - 11, 17, 85
DSA - ES

User activities:
Navy - AS, OS, CG

Preparing activity:
Navy - EC

(Project 5910-0847-3)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:38Z
Check the source to verify that this is the current version before use.